

A New Paradigm in Drug Discovery

Multi-target Drugs (Small Molecules) for pain and CNS diseases

July 2026

Vivozon Group Organization



KOSDAQ LISTED COMPANY **OTC / ETC**
Formulation / Production / Sales



First-in-class & Best-in-class
Drug Discovery & Development

Vivozon Canada

Toronto,
J&J Innovation center

Global BD Center

Vivozon US

New Jersey

Global Clinical Center

Medical Writing / RA

- FDA IND / NDA
- FDA Corres.

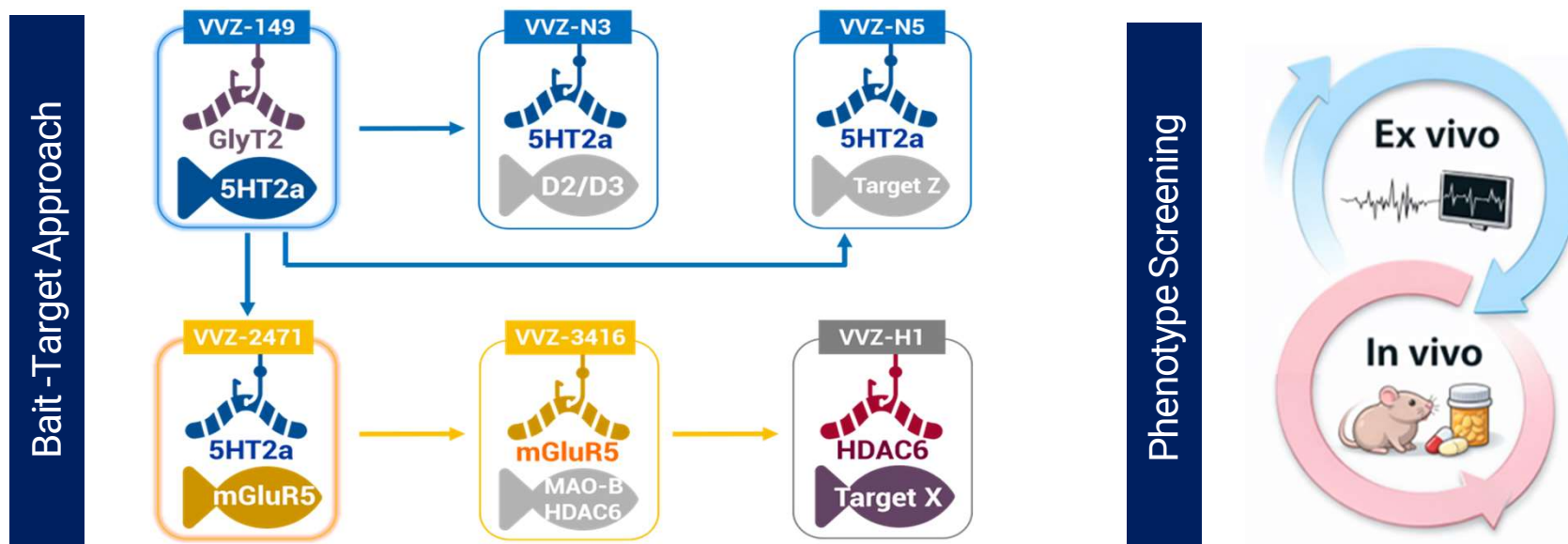


Vivozon is a biopharmaceutical company in pursuit of breakthrough innovation
for **pain and CNS therapeutics** with **multi-target drug discovery platform**

Multi-target Small Molecules for Pain and CNS Disorders

Pipeline

Project	VVZ-149 (Opiranserin) 		VVZ-2471	VVZ-3416	VVZ-N3	VVZ-N5	VVZ-H1
Indications (Route)	Postoperative Pain (IV) 	Acute Pain (Topical)	Neuropathic Pain, OUD (Oral)	Neurodegenerative disease (Oral)	Schizophrenia (Oral)	Antiplatelet Therapy (Oral)	Atrial fibrillation/ Cancer (Oral)
Development Stage	NDA (MFDA) Phase III (US)	Phase II completed	Phase II (KR) Phase Ib (US)	GLP-Tox	Preclinical Candidate	Lead Identified	Lead Identified



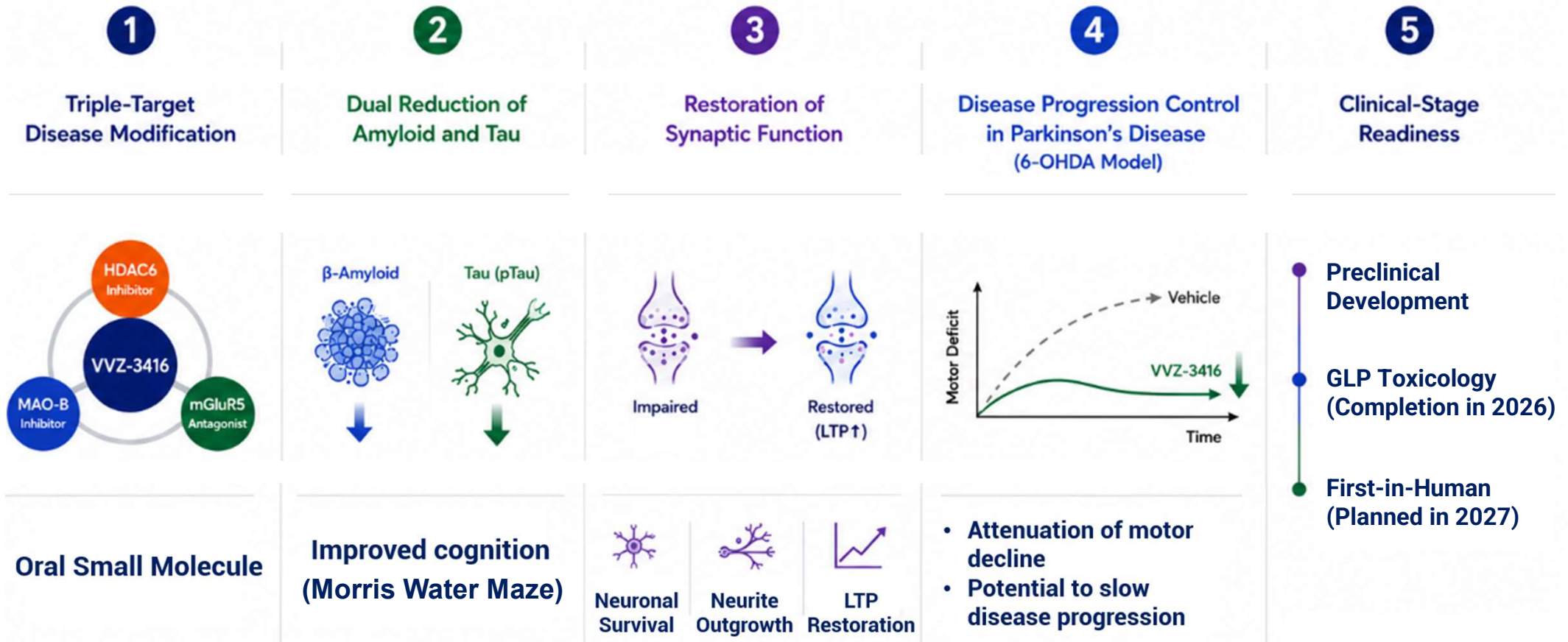
VVZ-3416

NEXT-GENERATION MULTI-TARGET DRUG

- **First-in-Class**
- **Alzheimer's & Parkinson's Disease**
- **Oral Formulation**
- **Current Status: GLP-Tox**

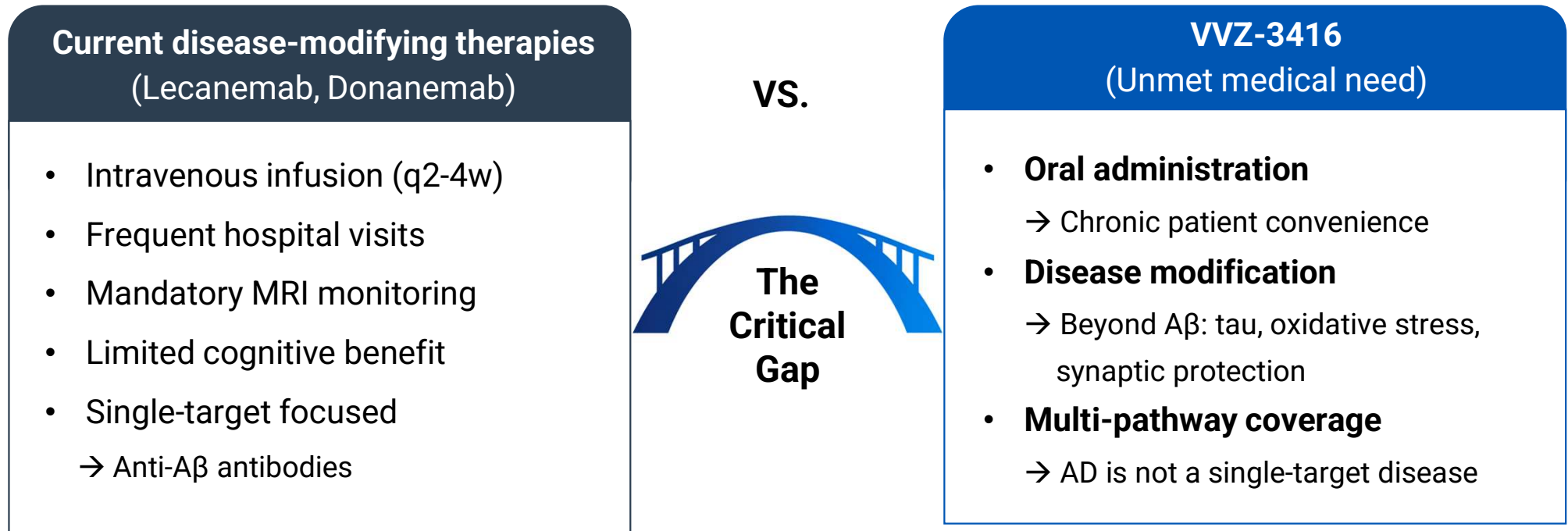
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Executive Summary



Alzheimer's Disease Treatment Landscape

Bridging the critical gap for oral, chronic disease-modifying therapy



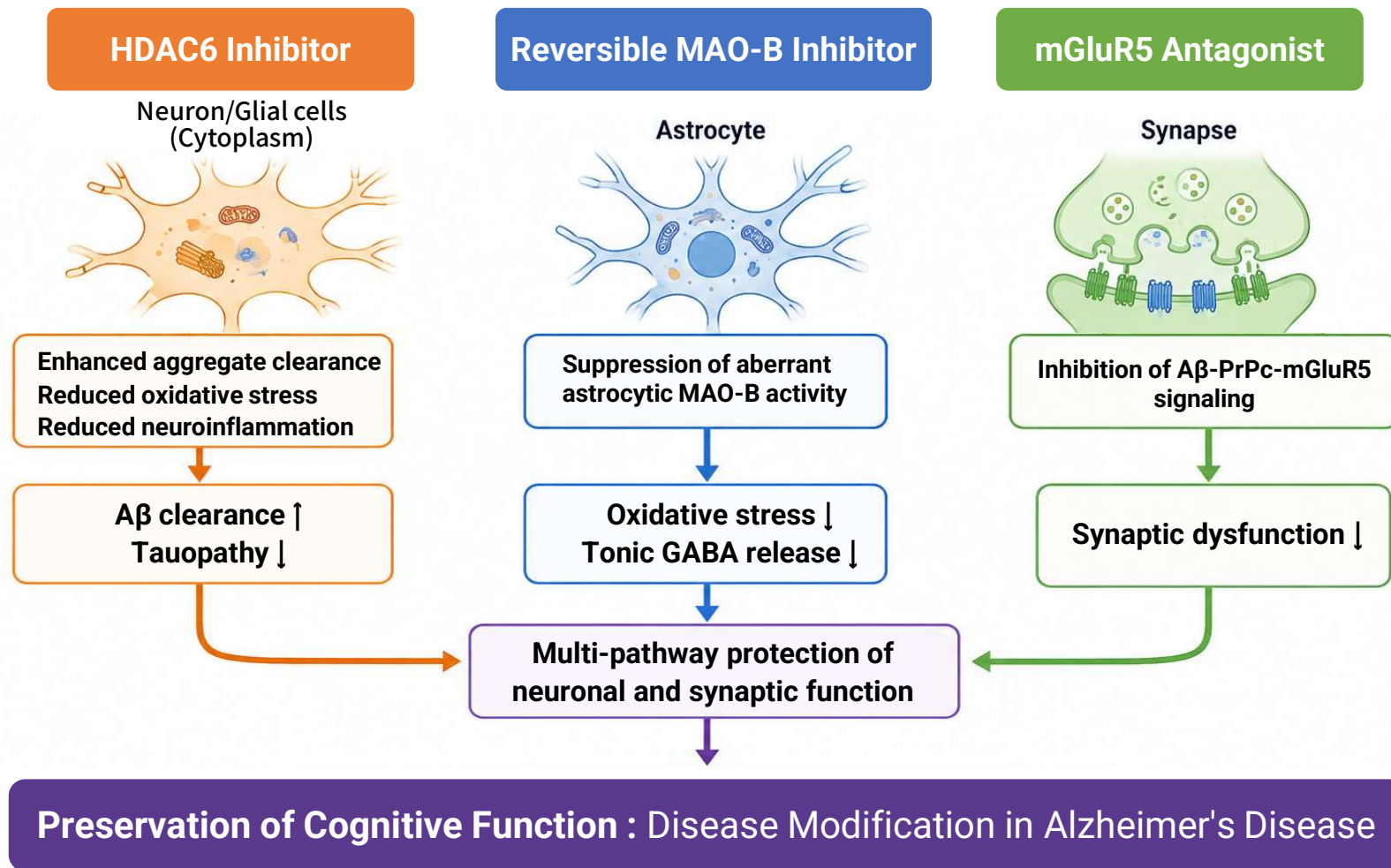
VVZ-3416 is designed to address this gap:

An oral, multi-target, CNS small molecule for chronic disease modification

Multi-Target Disease Modification Strategy for Alzheimer's Disease

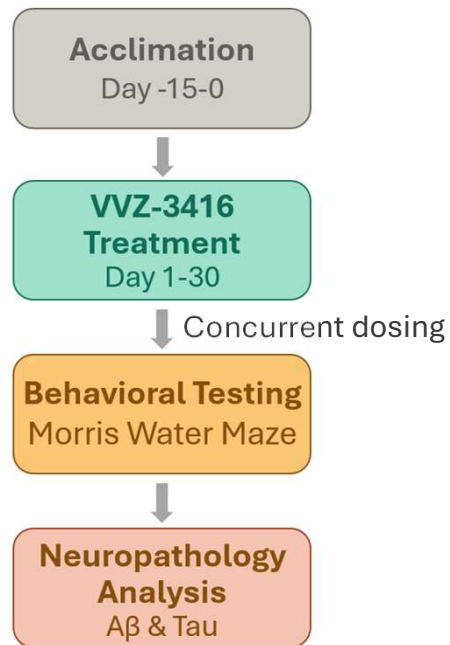
VVZ-3416

**Triple-Target
Small
Molecule**

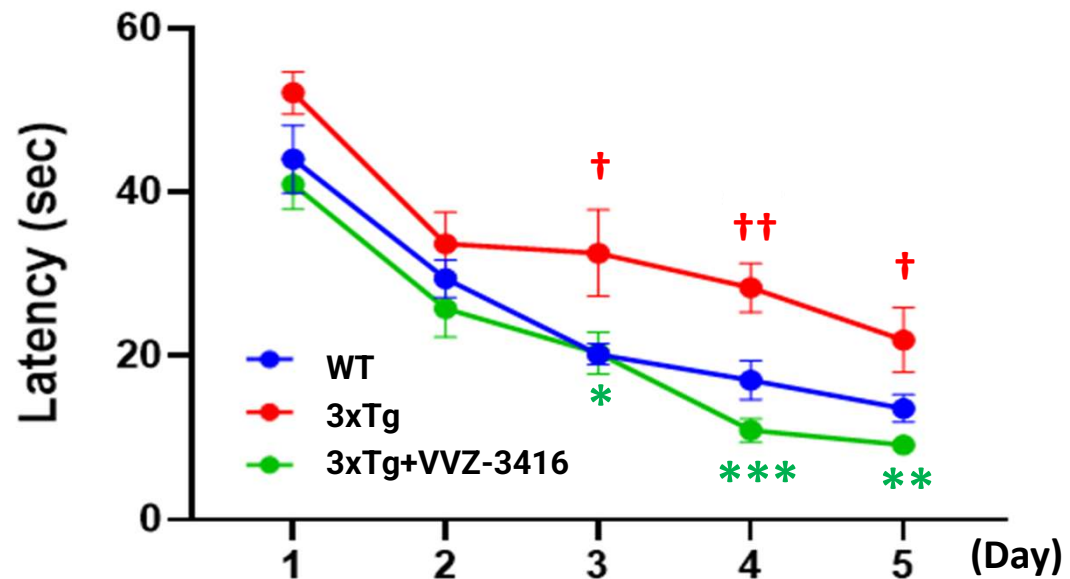


VVZ-3416 Improved Cognitive Function in 3xTg-AD Mice

❖ Procedure



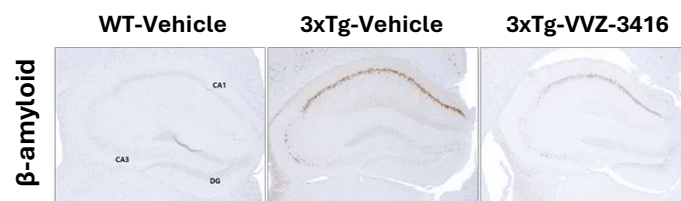
❖ Morris Water Maze Test



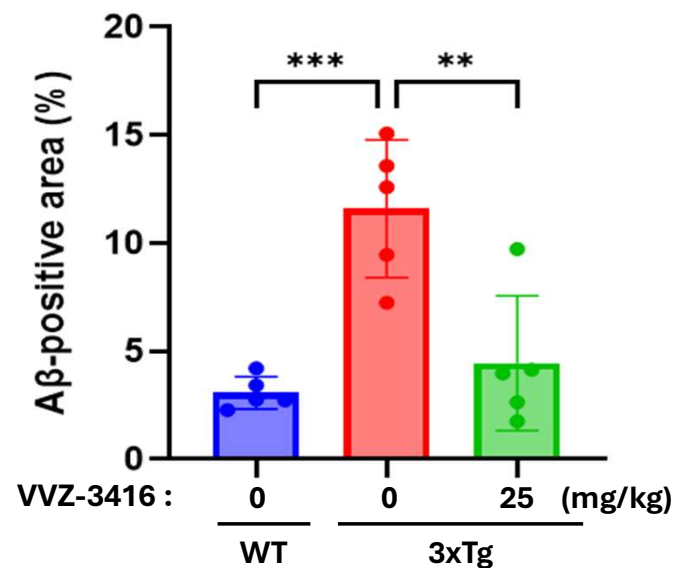
Statistical analysis was performed using two-way repeated measures ANOVA followed by Tukey's post hoc test. †p<0.05, ††p<0.01, †††p<0.001 vs. G1 (WT); *p<0.05, **p<0.01, ***p<0.001 vs. G2 (TG vehicle) (n=10/group)

Behavioral findings suggest cognitive benefits of VVZ-3416

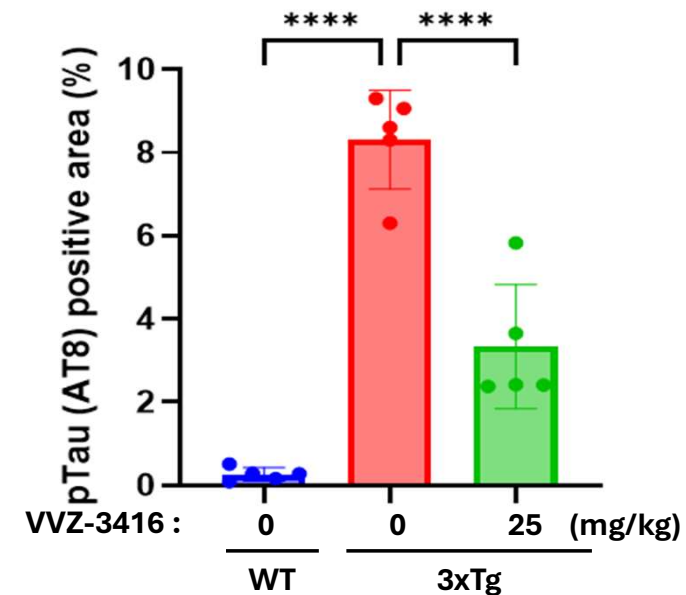
VVZ-3416 Reduced Both Amyloid and Tau Pathology in 3xTg-AD Mice



- Hippocampus: CA1



- Hippocampus: CA1



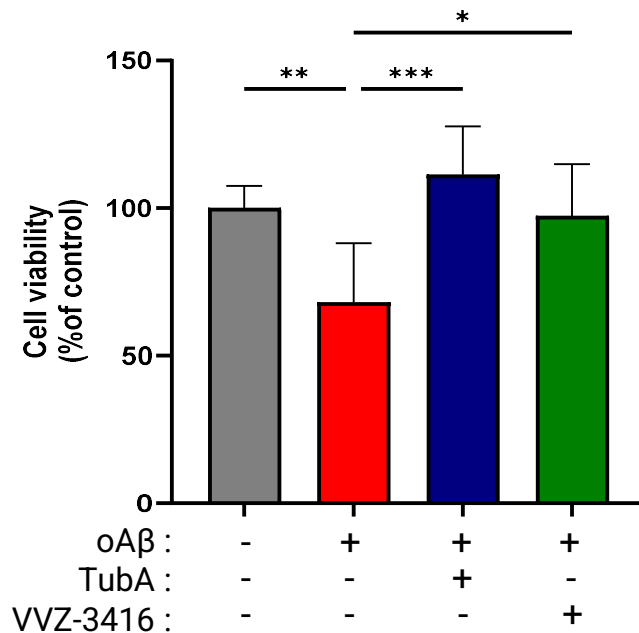
p<0.01, *p<0.001, ****p<0.0001

Reduced amyloid and tau pathology supports the disease-modifying potential of VVZ-3416

VVZ-3416 Supports Neuronal Survival and Neurite Outgrowth

HDAC6

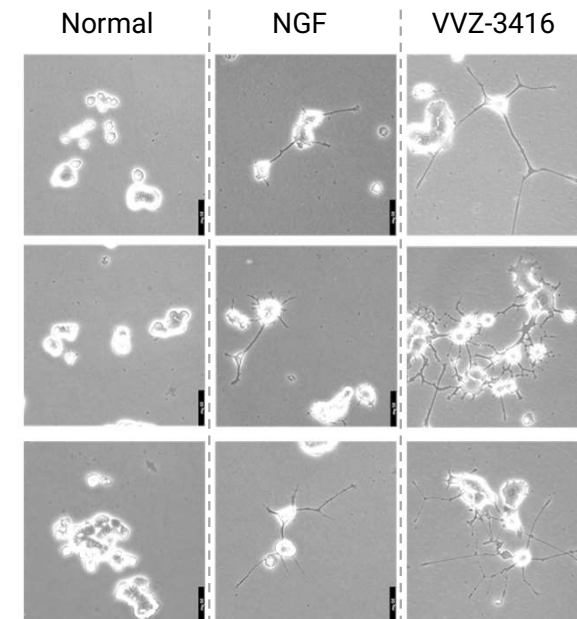
❖ Protection against oA β -induced cytotoxicity (SH-SY5Y)



• oA β : oligomeric amyloid beta, TubA: tubastatin A (HDAC6 inhibitor)

*p<0.05, **p<0.01, ***p<0.001, n=6/group

❖ Promotion of neurite outgrowth (PC12, 120 h)



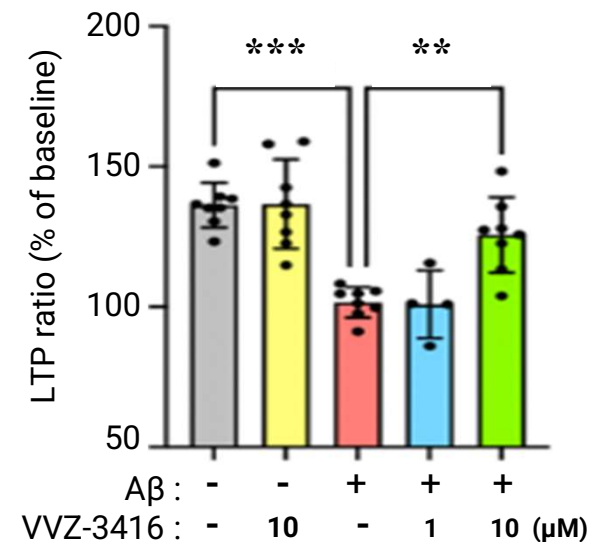
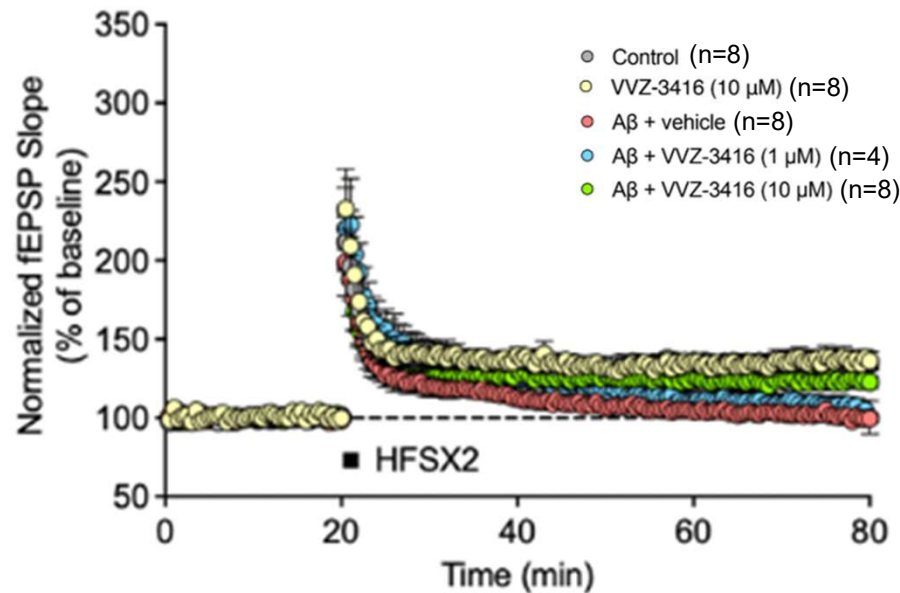
• NGF: nerve growth factor

VVZ-3416 supports both neuronal survival and neurite extension in cellular models

VVZ-3416 Supports Synaptic Function Beyond Neuronal Survival

mGluR5

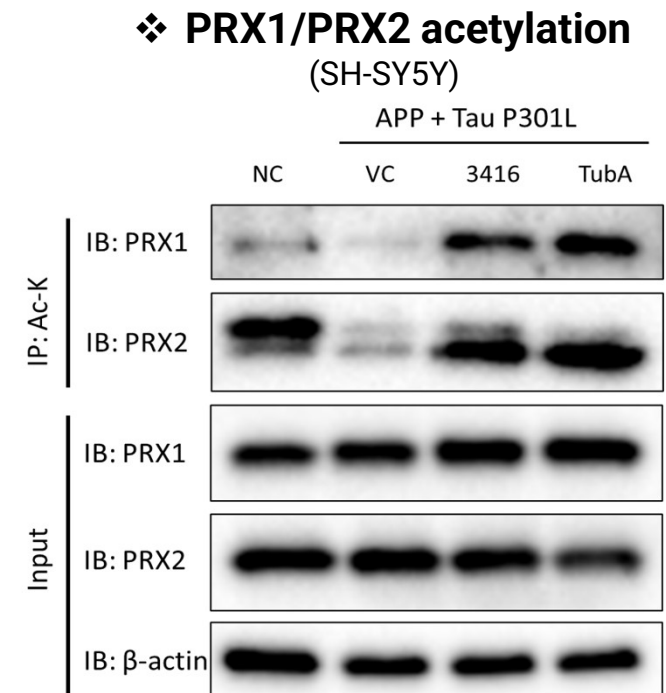
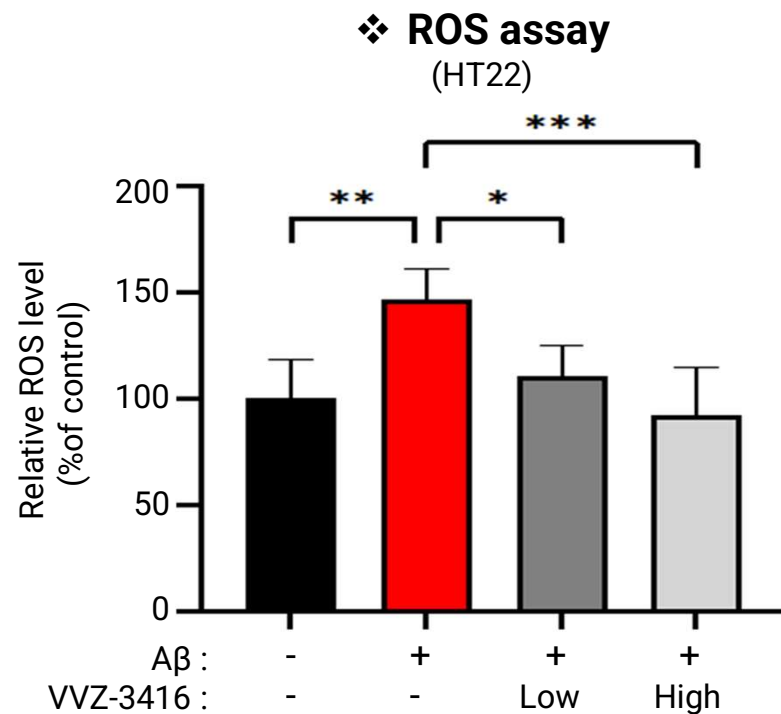
❖ Restoration of A β -impaired LTP



Improved LTP demonstrates functional preservation of synaptic networks beyond neuronal survival and neurite extension.

VVZ-3416 Mitigates Oxidative Stress Through Redox Regulation

MAO-B

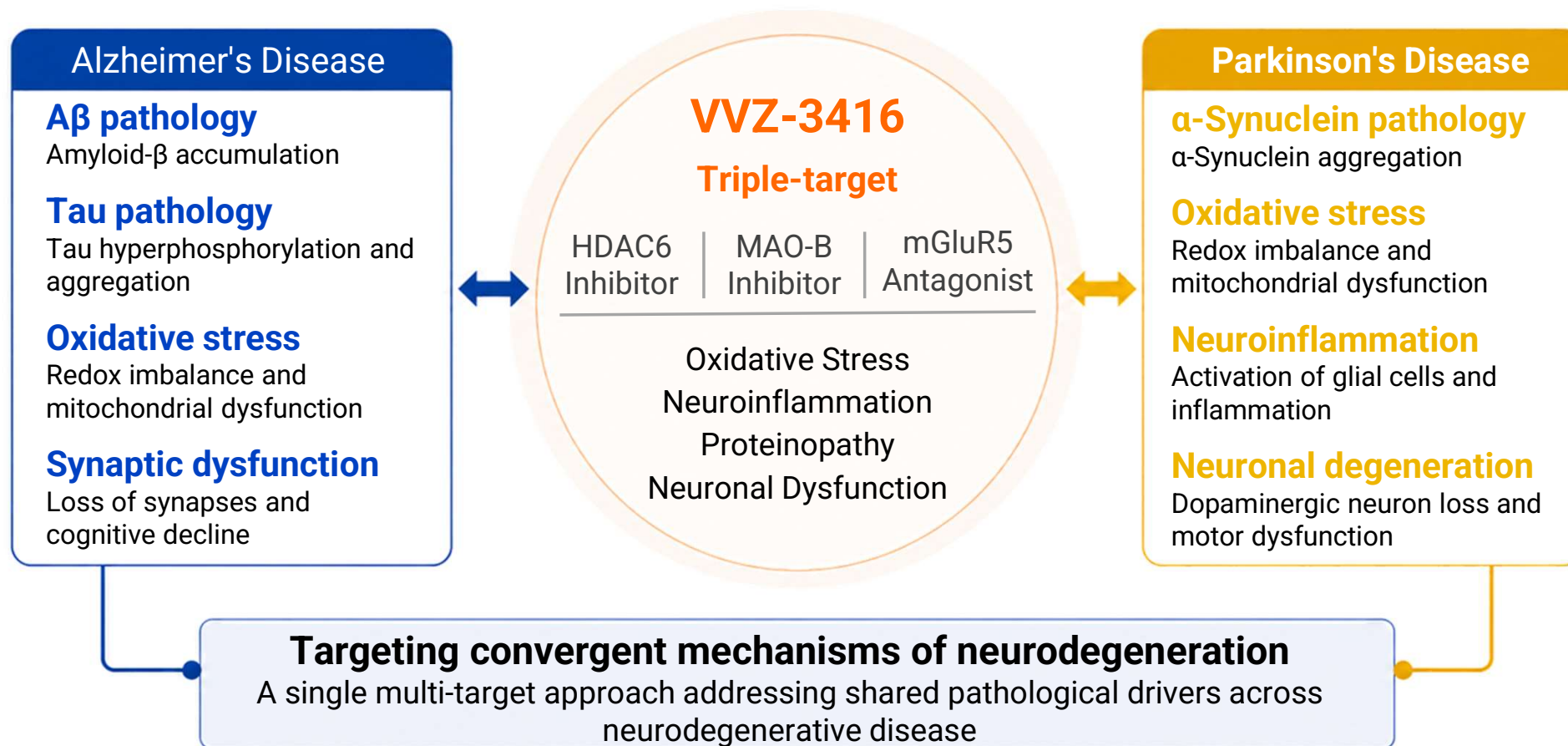


- HT22: Mouse hippocampal neuronal cell line, APP: Amyloid Precursor Protein; Tau P301L: pathogenic tau mutant; ROS: Reactive Oxygen Species; PRX1/PRX2: Peroxiredoxin 1/2 antioxidant enzymes; Ac-K : Acetyl-Lysine; TubA: Tubastatin A (selective HDAC6 inhibitor).

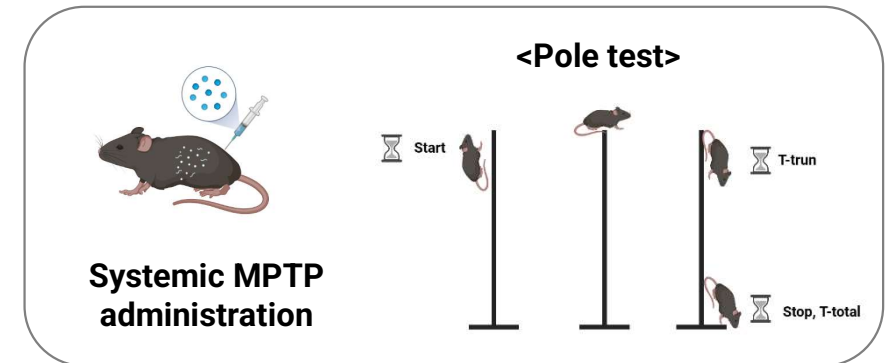
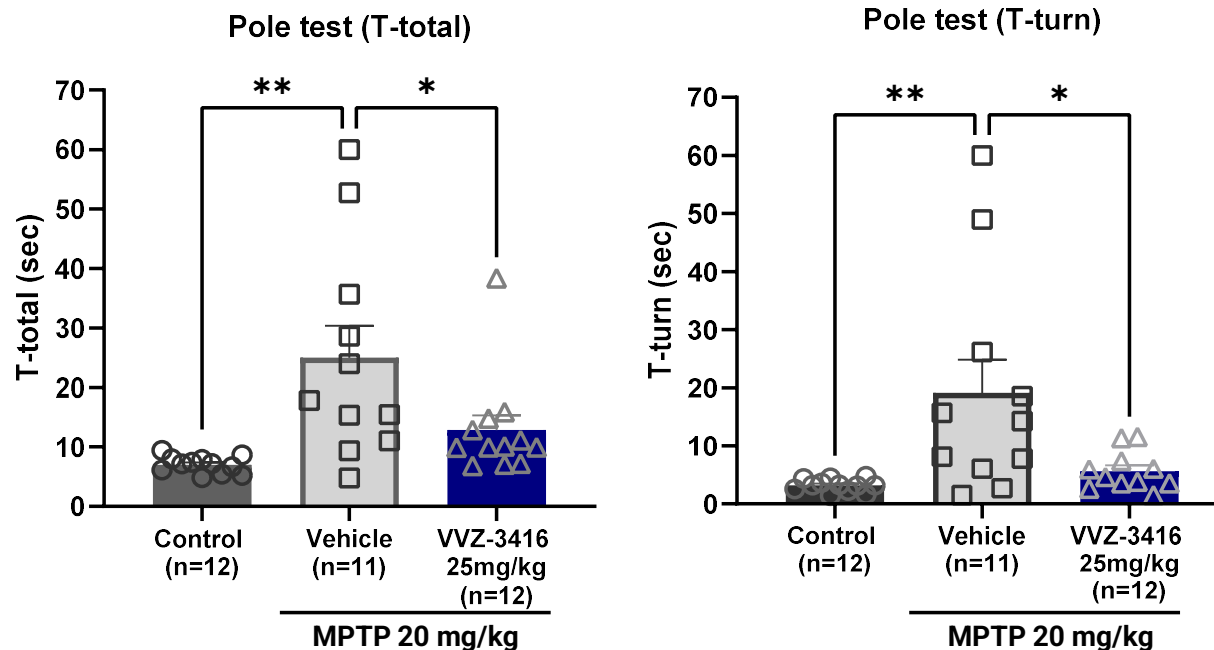
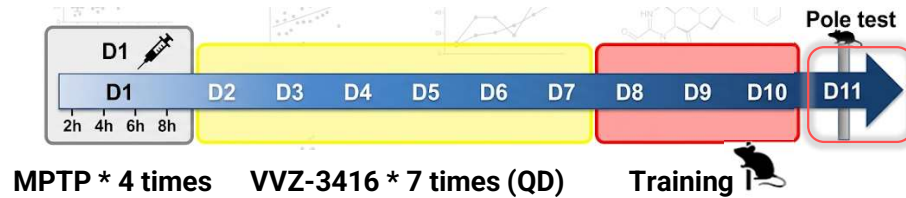
*p<0.05, **p<0.01, ***p<0.001, n=5-6/group

VVZ-3416 reduces oxidative stress through restoration of cellular redox homeostasis

Shared Pathological Drivers Across Neurodegenerative Diseases



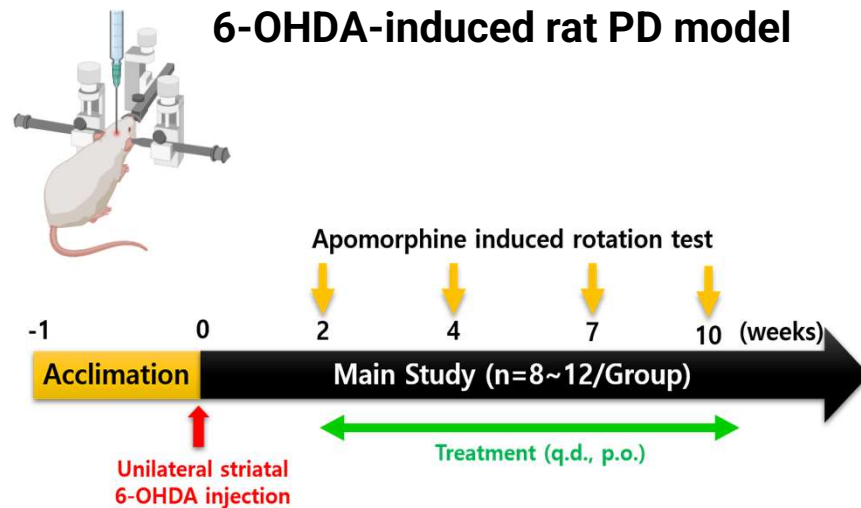
In Vivo Efficacy : MPTP-induced Mouse PD Model



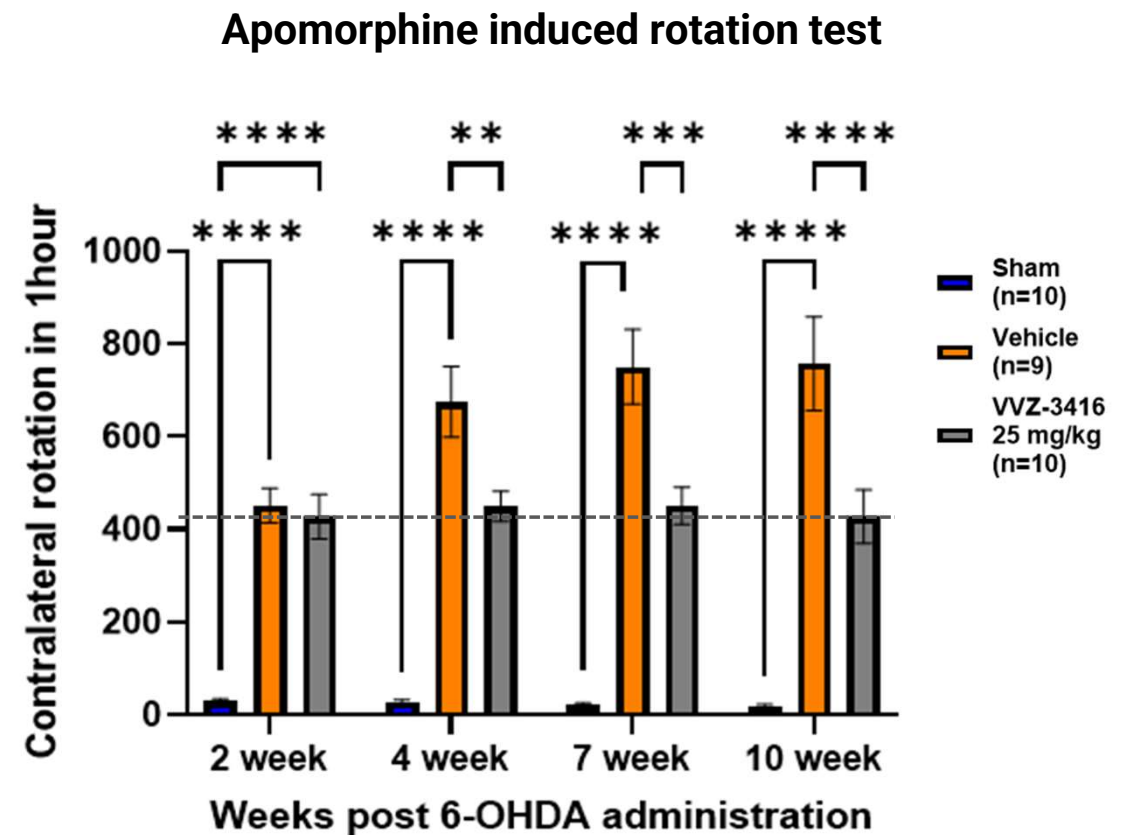
✓ **VVZ-3416 significantly reversed MPTP-induced motor deficits, as assessed by the pole test**

*p<0.05, **p<0.01

In Vivo Efficacy: 6-OHDA-induced Rat PD model



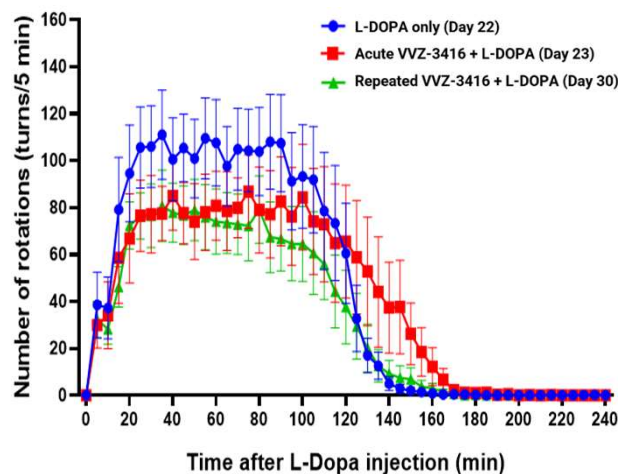
“Prevention of disease progression”



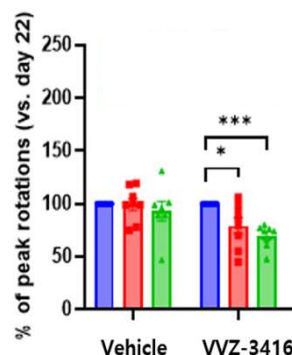
p<0.01, *p<0.001, ****p<0.0001 (n=8-12/group)

Exploratory Signal: PD-LID

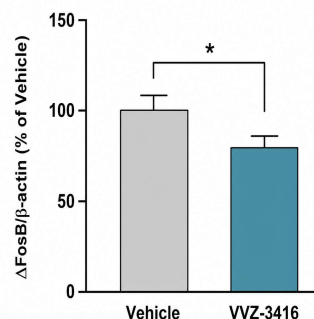
❖ 6-OHDA-Lesioned Rat Model of L-DOPA-Induced Dyskinesia



Peak rotations



Striatal Δ FosB



VVZ-3416 25 mg/kg p.o. administered as a single dose on Day 23 and following 7 days of repeated dosing on Day 30. Striatal tissue was subsequently collected for Δ FosB (n=8/group).

Reduced peak rotational response and striatal Δ FosB while preserving antiparkinsonian efficacy

ON-TIME: No Change

Motor-effective window preserved 6(single & repeat)

PEAK ROTATION: ↓ Reduced

Significant decrease vs. baseline (p<0.05, both regimens)

STRIATAL Δ FosB: ↓ Reduced

Chronic LID biomarker; significantly lower vs. vehicle (p<0.05)

Current Limitation: AIMs score not yet assessed; reduced peak response has not been dissociated from motor sensitivity change.

Profile summary



VVZ-3416

Triple target

mGluR5 – negative allosteric modulator

HDAC6 – highly selective inhibitor

MAO-B – selective & reversible inhibitor

Target

- mGluR5 IC₅₀ 0.559 µM / HDAC6 IC₅₀ 0.101 µM / MAO-B IC₅₀ 0.76 µM
- No off-target signal: GPCR panel screen (168 receptors)

DMPK

- High oral bioavailability: rat ~80%
- High liver microsomal stability across species (≥80%)
- High BBB penetration: brain/plasma ratio ~1.0 (rat)

CYP (metabolite)

- CYP1A1 inhibition IC₅₀ 3.35 µM
- Primarily metabolized via CYP1A1

Tox

- Rat 4-week repeated-dose toxicity study: completed
- Monkey 4-week repeated-dose toxicity study: dosing completed

Status

- Monkey GLP toxicology study: dosing completed; report in preparation
- Drug substance (DS) manufacturing: in progress
- IND submission: planned Q2 2027

Thank you